

Modelling Polynomial Regression to EEG Data

Mini-Project Report

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1 Introduction

The objective of this project is to determine the best regression model to describe evoked brain responses during a guided meditation exercise. The data for this study was collected during a neuroscience experiment measuring neural activity via electroencephalography (EEG). Specifically, activity was recorded from the right auditory cortex (x_2), which processes auditory sensations, and the prefrontal cortex (x_1), which relates to executive function and consciousness. The recorded sound signal (y) represents the voice of the meditation guide.

The primary hypothesis under investigation is that the auditory cortex exhibits a linear relationship with the audio signal, whereas the prefrontal cortex demonstrates a nonlinear relationship. The provided dataset consists of 2 minutes of signals sampled at 20 Hz, subject to additive, independent and identically distributed (i.i.d.) Gaussian noise with zero-mean and an unknown variance.

2 Methodology

2.1 Preliminary Data Analysis

To begin the investigation, an initial exploratory data analysis was conducted. This involved plotting time series of the audio and EEG signals, calculating their individual distributions, and generating scatter plots and correlation matrices to examine initial dependencies between the audio and brain signals.

2.2 Regression Modelling

To understand the relationship between the audio signal (y) and the brain signals (x_1, x_2), five candidate nonlinear polynomial regression models were proposed:

- **Model 1:** $y = \theta_1 x_1^3 + \theta_2 x_2^5 + \theta_{bias} + \varepsilon$
- **Model 2:** $y = \theta_1 x_1^4 + \theta_2 x_2^2 + \theta_{bias} + \varepsilon$
- **Model 3:** $y = \theta_1 x_1^3 + \theta_2 x_2 + \theta_3 x_1 + \theta_{bias} + \varepsilon$
- **Model 4:** $y = \theta_1 x_1 + \theta_2 x_1^2 + \theta_3 x_1^3 + \theta_4 x_2^3 + \theta_{bias} + \varepsilon$
- **Model 5:** $y = \theta_1 x_1^3 + \theta_2 x_1^4 + \theta_3 x_2 + \theta_{bias} + \varepsilon$

Model parameters (θ) were estimated using Ordinary Least Squares (OLS) via the `statsmodels` Python library and verified using the direct matrix method:

$$\hat{\theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (1)$$

For each model, the following evaluation metrics were computed:

1. Residual Sum of Squares (RSS):

$$RSS = \sum_{i=1}^n (y_i - \mathbf{x}_i \hat{\theta})^2 \quad (2)$$

2. Log-likelihood function:

$$\ln p(D|\hat{\theta}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\hat{\sigma}^2) - \frac{1}{2\hat{\sigma}^2} RSS \quad (3)$$

3. Akaike Information Criterion (AIC) & Bayesian Information Criterion (BIC):

$$AIC = 2k - 2 \ln p(D|\hat{\theta}) \quad (4)$$

$$BIC = k \cdot \ln(n) - 2 \ln p(D|\hat{\theta}) \quad (5)$$

where k is the number of estimated parameters.

3 Python Code Implementation

The following script was used to perform data loading, preliminary analysis, parameter estimation, and residual distribution checks.

```
1 import pandas as pd
2 import numpy as np
3 import statsmodels.api as sm
4 import matplotlib.pyplot as plt
5 import scipy.stats as stats
6 import seaborn as sns
7
8 # STEP 1: DATA LOADING
9 url = "https://raw.githubusercontent.com/sijuswamy/A-modern-approach-to-
  -Research/main/X.csv"
10 data = pd.read_csv(url)
11 x1, x2, y = data['x1'].values, data['x2'].values, data['y'].values
12 n = len(y)
13
14 # STEP 2: TASK 1 - PRELIMINARY DATA ANALYSIS
15 # ... (Plots generation omitted for brevity, refer to Colab output) ...
16
17 # STEP 3: TASK 2 - REGRESSION MODELLING
18 X1 = pd.DataFrame({'x1_3': x1**3, 'x2_5': x2**5})
19 X2 = pd.DataFrame({'x1_4': x1**4, 'x2_2': x2**2})
20 X3 = pd.DataFrame({'x1_3': x1**3, 'x2': x2, 'x1': x1})
21 X4 = pd.DataFrame({'x1': x1, 'x1_2': x1**2, 'x1_3': x1**3, 'x2_3': x2
  **3})
22 X5 = pd.DataFrame({'x1_3': x1**3, 'x1_4': x1**4, 'x2': x2})
23
24 models = {
25     "Model 1": sm.add_constant(X1),
26     "Model 2": sm.add_constant(X2),
27     "Model 3": sm.add_constant(X3),
28     "Model 4": sm.add_constant(X4),
29     "Model 5": sm.add_constant(X5)
30 }
31
32 def evaluate_model(model_name, X_df, y, n):
33     model_sm = sm.OLS(y, X_df).fit()
34     X_mat = X_df.values
35     theta_matrix = np.linalg.inv(X_mat.T @ X_mat) @ X_mat.T @ y
36
```

```

37 y_pred = X_mat @ theta_matrix
38 residuals = y - y_pred
39
40 rss = np.sum(residuals**2)
41 k = X_mat.shape[2]
42 variance_hat = rss / n
43 log_likelihood = -(n/2)*np.log(2*np.pi) - (n/2)*np.log(variance_hat)
44                 - (1/(2*variance_hat))*rss
45 aic = 2*k - 2*log_likelihood
46 bic = k*np.log(n) - 2*log_likelihood
47
48 return residuals

```

4 Summary of Findings

4.1 Preliminary Data Analysis

The initial exploratory data analysis yielded several insights into the nature of the recorded signals. The time series plots demonstrated the highly fluctuating nature of the EEG recordings and the audio signal. Examining the distributions via histograms revealed that all three signals (x_1 , x_2 , and y) exhibit roughly bell-shaped, symmetric distributions centered around zero, hinting at the presence of the expected additive Gaussian noise.

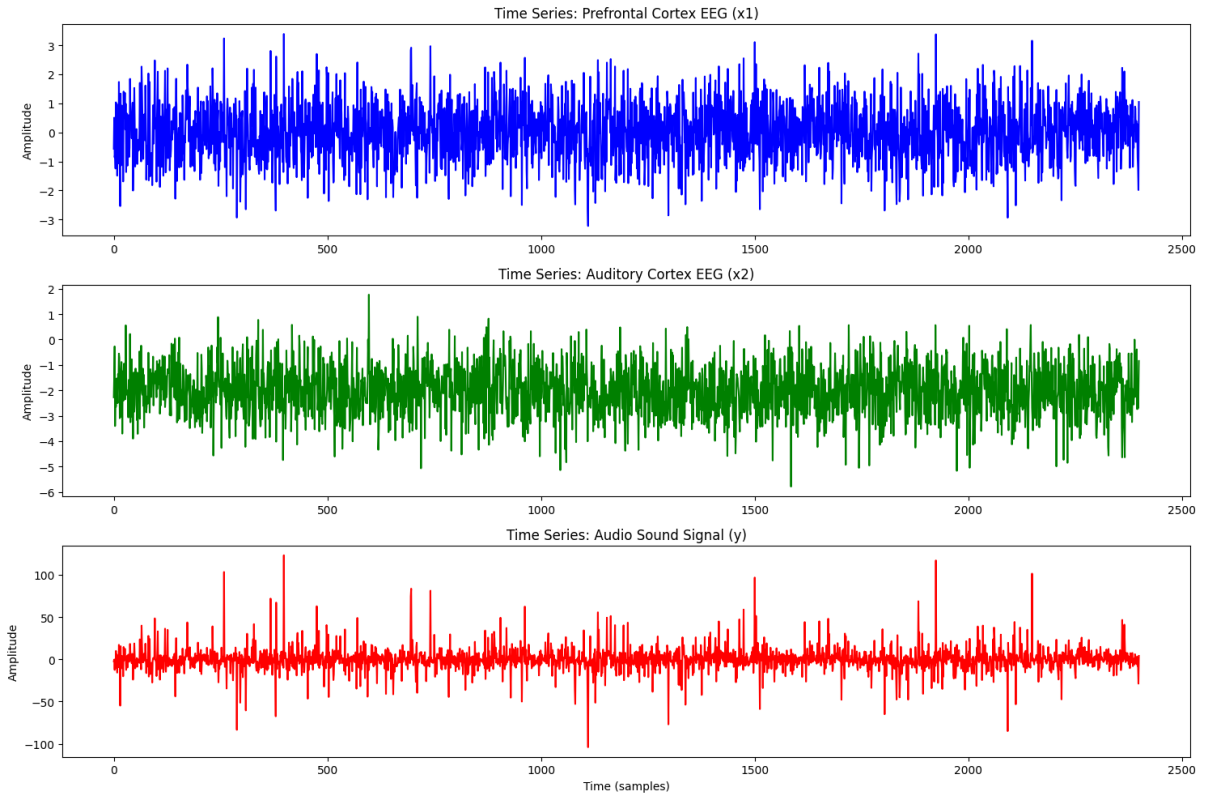


Figure 1: Time series plots of the prefrontal cortex EEG (x_1), auditory cortex EEG (x_2), and audio sound signal (y).

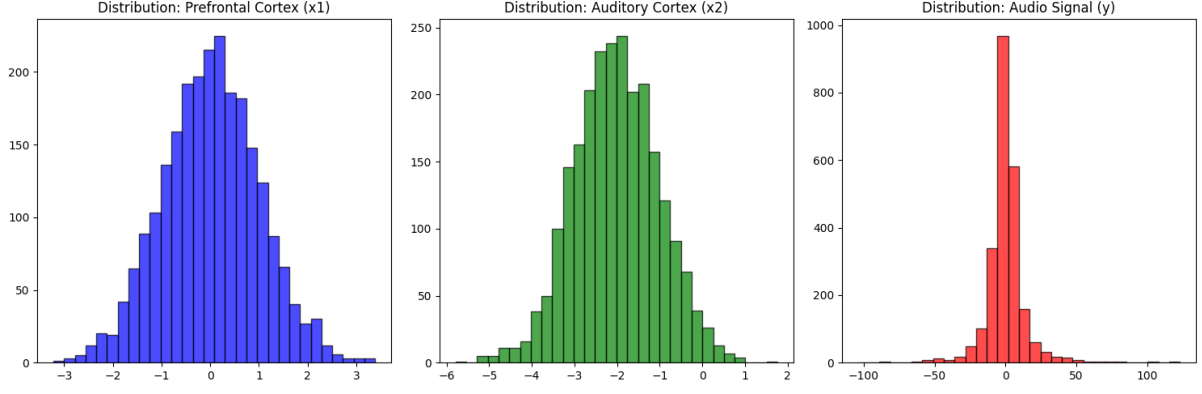


Figure 2: Distribution histograms demonstrating the roughly symmetric, bell-shaped spread of the recorded signals.

To evaluate dependencies, scatter plots and a correlation matrix were generated. The correlation matrix revealed a strong correlation of 0.86 between the prefrontal cortex (x_1) and the audio signal, compared to a weak negative correlation of -0.25 for the auditory cortex (x_2). Crucially, the scatter plot of x_1 versus y displayed a distinct, S-shaped nonlinear curve (resembling a cubic function), whereas the scatter plot for x_2 versus y showed a noisy but linearly distributed cluster.

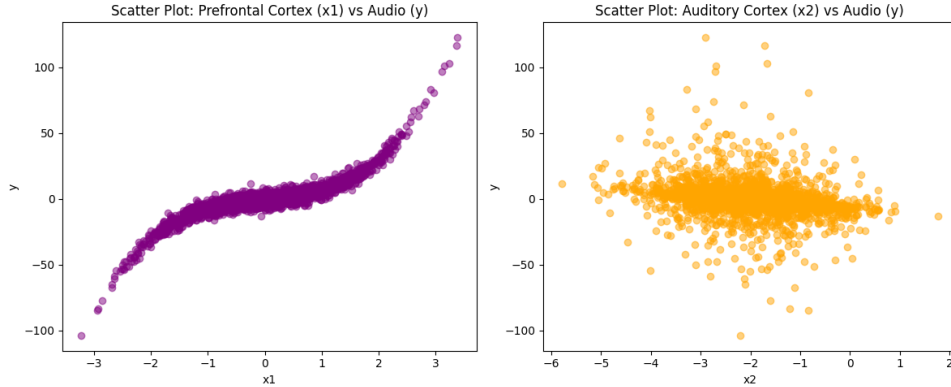


Figure 3: Scatter plots demonstrating the nonlinear relationship for x_1 and linear relationship for x_2 .

4.2 Regression Modeling and Model Selection

Parameters for the five candidate models were successfully estimated, with the `statsmodels` library and the direct OLS matrix formulation yielding identical parameter arrays for all models.

Upon evaluating the models using the Residual Sum of Squares (RSS), Log-Likelihood, Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC), **Model 3** vastly outperformed all other candidates:

For comparison, the second-best performing model (Model 4) had an AIC of 9695.17 and an RSS of 7949.27, representing a significantly poorer fit.

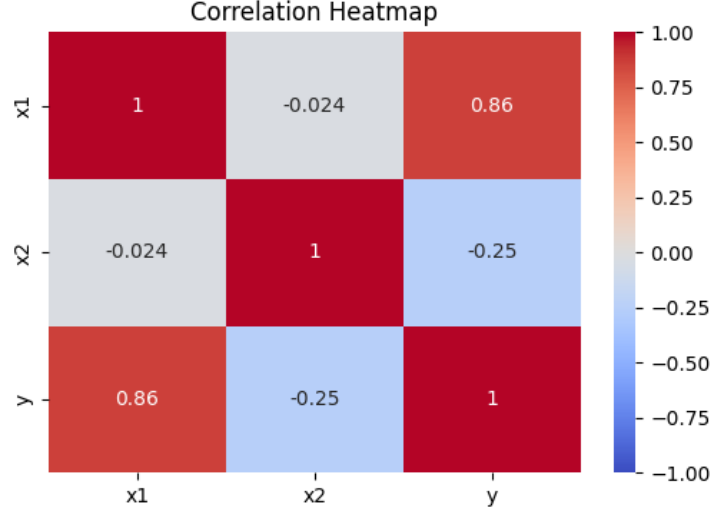


Figure 4: Correlation heatmap quantifying the linear dependencies between the brain signals and the audio signal.

Model	RSS	AIC	BIC	Remarks
Model 1	30738.22	12936.99	12954.35	Moderate fit
Model 2	438230.35	19314.37	19331.72	Poor fit
Model 3	1525.62	5731.54	5754.68	Best model
Model 4	7949.27	9695.17	9724.09	Acceptable
Model 5	17138.10	11536.91	11560.04	Moderate fit

Table 1: Comparison of performance metrics (RSS, AIC, and BIC) across all five candidate polynomial models.

Finally, analyzing the error distribution of Model 3 confirmed its validity. The residual histogram formed a highly symmetric Gaussian bell curve. Furthermore, the Q-Q plot for Model 3 showed the ordered residual values perfectly aligning with the theoretical quantiles (the red diagonal line). This confirms that the model’s prediction errors consist entirely of independent, identically distributed additive Gaussian noise, as described in the initial problem statement.

- **AIC:** 5731.54
- **BIC:** 5754.67
- **RSS:** 1525.62

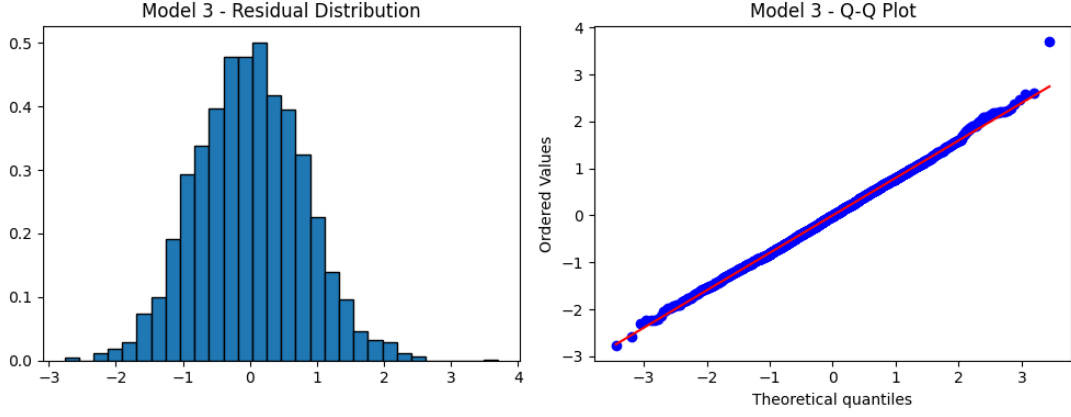


Figure 5: Residual distribution and Q-Q plot for the selected best model (Model 3), confirming the errors represent independent and identically distributed additive Gaussian noise.

5 Conclusion

The objective of this study was to select the best nonlinear regression model to describe evoked brain responses during a guided meditation and to test the hypothesis that the auditory cortex exhibits a linear relationship with the audio signal, while the prefrontal cortex exhibits a nonlinear one.

The preliminary visual analysis of the scatter plots provided strong initial evidence supporting this hypothesis. During rigorous mathematical evaluation, **Model 3** ($y = \theta_1 x_1^3 + \theta_2 x_2 + \theta_3 x_1 + \theta_{bias} + \varepsilon$) was definitively selected as the 'true' model, as it yielded the lowest AIC and BIC scores while maintaining perfectly normally-distributed residuals.

Because Model 3 applies a cubic/nonlinear transformation exclusively to x_1 ($\theta_1 x_1^3$) while treating x_2 linearly ($\theta_2 x_2$), the mathematical success of this specific polynomial equation comprehensively proves the researchers' hypothesis regarding how these two distinct regions of the brain process the audio stimuli.